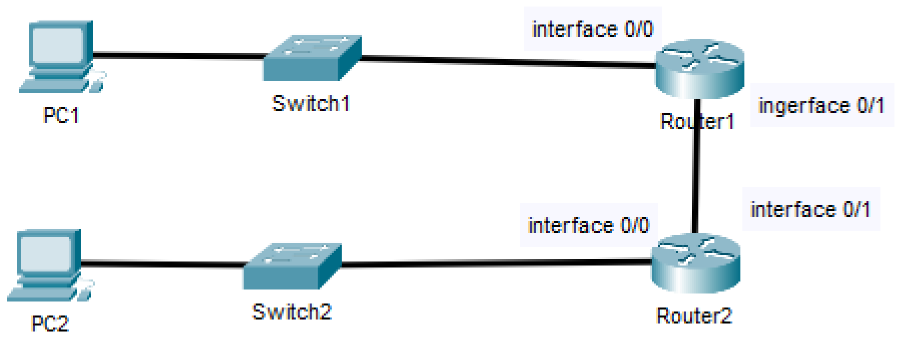
Lab 4: Building a Switch and Router Network on both IPv4 and IPv6

Topology



Step 0: useful commands

Router#show ip int brief	show a list of the routers network interfaces. You can find the names and status of all interfaces here
Router#write erase	delete the startup config
Router#reload	reboot the router
Router(config)#interface <NameOfInterface>	navigate to the configuration mode of an interface
Router(config-if)#ip address <IPAddress> <Subnetmask>	Set the IP address and subnet mask of a router's interface
Router(config-if)#no shutdown	Enable the interface
Router(config)#ip route <destination network address> <destination network subnet mask> <ip address of next router>	Configure a static route
Router#show ip route	Show router's routing table
Router#show interface Gigabitethernet 0/0	Show information of a specific router interface
Router(config)#ipv6 unicast-routing	Enable SLAAC on the router

FOR THIS LAB, YOU CAN WORK ALONE OR IN A GROUP OF 2.

Step 1: Subnet the network

To randomize the IP addresses that each group uses, you will use the last 2 digits of your student number in the IP network address that you can use for your network (if you work in a group, choose one student number). The network that you can use to subnet the network is 10.10.(student card digits).0/22.

e.g. for the student number r215468, the network address becomes 10.10.68.0/22

We will use: 10.10.9.0/22 (my student number is r1057809)

Range: 10.10.8.0 - 10.10.11.255

Usable: 10.10.8.1 - 10.10.11.254

LAN 1	10.10.8.0/23	255.255.254.0
LAN 2	10.10.10.0/24	255.255.255.0
Router to Router link	10.10.11.252/30	255.255.255.252

LAN 1 Range: 10.10.8.0 - 10.10.9.255

LAN 2 Range: 10.10.10.0 - 10.10.10.255

Router link: 10.10.11.252 - 10.10.11.255

Use this address to subnet the network given above and assign IP addresses to each device in the network. Make sure to maximize the number of IP addresses on each network.

Let the table be checked before going to the next step

Assigned IP addresses:

Device	Interface	IP Address	Subnet Mask	Default Gateway (if applicable)
Router1	0/0	10.10.8.1	255.255.254.0	-
Router1	0/1	10.10.11.253	255.255.255.252	-
Router2	0/0	10.10.10.1	255.255.255.0	-
Router2	0/1	10.10.11.254	255.255.255.252	-
PC1	NIC	10.10.8.2	255.255.254.0	10.10.8.1
PC2	NIC	10.10.10.2	255.255.255.0	10.10.10.1

Step 2: Prepare your network in packet tracer

- Before building a network on physical devices, it is always interesting to simulate your network first. Building a virtual network is quicker, you can find problems easier (in simulation mode) and this way you can prepare all the commands that you need in the physical network that you know will work.
- Build the network as shown in the image above

IPv4

- Assign IP addresses to the PC's, switches and routers.
- Before testing the network connectivity, think about how PC's and routers work and predict which devices can reach each other with the current configuration.
- Test network connectivity between all the devices and see if your prediction was correct. You can also use the simulation mode to see what is happening.
- PC1 cannot reach PC2, PC2 cannot and PC1. The reason is that the subnet of PC2 is not yet known to Router1 and the subnet of PC1 is not yet known to Router2. Have a look at the routing table of both routers to confirm.
- Configure a static route on Router1 to the subnet of PC2, and a static route on Router2 to the subnet of PC1. Confirm the configuration in the router's routing table.
- Test network connectivity between PC1 and PC2.

Router interface configuration (IPv4)

Router 1

```
Router> enable

Router# configure terminal

Router(config)# hostname R1

R1(config)# interface g0/0

R1(config-if)# ip address 10.10.8.1 255.255.254.0

R1(config-if)# no shutdown

R1(config-if)# exit

R1(config)# interface g0/1

R1(config-if)# ip address 10.10.11.253 255.255.255.252

R1(config-if)# no shutdown

R1(config-if)# exit
```

Router 2

```
Router> enable

Router# configure terminal

Router(config)# hostname R2

R2(config)# interface g0/0

R2(config-if)# ip address 10.10.10.1 255.255.255.0

R2(config-if)# no shutdown

R2(config-if)# exit

R2(config)# interface g0/1

R2(config-if)# ip address 10.10.11.254 255.255.255.252

R2(config-if)# no shutdown
```

Static routes (IPv4)

Router 1

```
R1(config)# ip route 10.10.10.0 255.255.255.0 10.10.11.254
```

Router 2

```
R2(config)# ip route 10.10.8.0 255.255.254.0 10.10.11.253
```

IPv6

- Only PC1 and Router1 have to be configured with IPv6. In IPv6, you don't need to configure static IP addresses to the PC's. SLAAC (Stateless Address Autoconfiguration) enables PC's connected to an IPv6 router to automatically configure an IP address in the correct network.
- Configure the IPv6 address 2001:A:CAFE:1::1/64 to Router1 and enable IPv6 unicast routing.
- PC1 should automatically get an IPv6 address
- Test network connectivity between PC1 and Router1 over IPv6

IPv6 SLAAC configuration

```
R1(config)# ipv6 unicast-routing

R1(config)# interface g0/0

R1(config-if)# ipv6 address 2001:A:CAFE:1::1/64
```

Verification

`show ip interface brief`	Controleer of alle poorten op 'up/up' staan.
`show ip route`	Controleer of de statische routes (gemarkeerd met 'S') aanwezig zijn.
`show ipv6 interface brief`	Verifieer de IPv6 status en het link-local adres.
`show arp`	Bekijk de Layer 2 naar Layer 3 mapping (MAC naar IP).

Step 3: Building the physical network

- Cable the physical network as shown in the topology
- During this lab, you will only have to configure the router. However, you should reset your switches and routers first.

IPv4

- Configure the network for IPv4
  - Assign names to the routers
  - Assign IP addresses to all the devices
  - Configure static routes on Router1 and Router2
- Test network connectivity between all the devices

From PC1 to PC2

```
PS C:\Users\Jaske> ping 10.10.10.2


Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time=2ms TTL=126

Reply from 10.10.10.2: bytes=32 time=2ms TTL=126

Reply from 10.10.10.2: bytes=32 time=2ms TTL=126

Reply from 10.10.10.2: bytes=32 time=3ms TTL=126


Ping statistics for 10.10.10.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
PS C:\Users\Jaske>
```

From PC2 to PC1

```
C:\Users\Adam>ping 10.10.8.2


Pinging 10.10.8.2 with 32 bytes of data:
Reply from 10.10.8.2: bytes=32 time=2ms TTL=126
Reply from 10.10.8.2: bytes=32 time=2ms TTL=126
Reply from 10.10.8.2: bytes=32 time=2ms TTL=126
Reply from 10.10.8.2: bytes=32 time=3ms TTL=126


Ping statistics for 10.10.8.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\Users\Adam>
```

IPv6

- Configure PC1 and Router1 for IPv6
    - Static IP address on Router1
    - Enable IPv6 unicast routing on Router1
    - PC1 should receive an IPv6 address automatically (SLAAC)
- Test network connectivity between PC1 and Router1

IPv6 in ipconfig

```
Ethernet adapter Ethernet:
   Connection-specific DNS Suffix  . : telenet.be
   Description . . . . . . . . . . . : Intel(R) Ethernet Controller I226-LM
   Physical Address. . . . . . . . . : FC-4C-EA-80-C5-A1
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2001:a:cafe:1:3543:acfa:f494:58ae(Preferred)
   Temporary IPv6 Address. . . . . . : 2001:a:cafe:1:5917:510d:9040:c6f4(Preferred)
   Link-local IPv6 Address . . . . . : fe80::26ee:e1be:ac9c:d22a%2(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.10.8.2(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.254.0
   Default Gateway . . . . . . . . . : fe80::c28c:60ff:fe3a:c1c0%2
                                       10.10.8.1
   DHCPv6 IAID . . . . . . . . . . . : 117198058
```

```
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-0B-49-6E-FC-4C-EA-80-C5-A1
   DNS Servers . . . . . . . . . . . : 2a02:1800:100::45:2
                                       2a02:1800:100::45:1
   NetBIOS over Tcpip. . . . . . . . : Enabled
   Connection-specific DNS Suffix Search List :
                                       telenet.be
```

ping Router 1 with its IPv6 address

```
PS C:\Users\Jaske> ping 2001:a:cafe:1::1

Pinging 2001:a:cafe:1::1 with 32 bytes of data:
Reply from 2001:a:cafe:1::1: time=1ms
Reply from 2001:a:cafe:1::1: time=1ms
Reply from 2001:a:cafe:1::1: time=1ms
Reply from 2001:a:cafe:1::1: time=1ms

Ping statistics for 2001:a:cafe:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
PS C:\Users\Jaske>
```

Step 4: Monitor the network for IPv4 traffic

- Start monitoring with Wireshark **<u>on both computers</u>** and ping over IPv4 from PC1 to PC2.
    - The Ping requests travel from PC1 to PC2. The same packet can be seen in Wireshark on both computers. On PC1, the ping request will leave the interface while the same packet will arrive in PC2. Fill in the table to see what changed in the packet

	Source MAC	Destination MAC	Source IP	Destination IP
Ping request (traveling from PC1 to PC2) as seen in Wireshark on PC1	fc:4c:ea:80:c5:a1 (from PC1)	c0:8c:60:3a:c1:c0 (Router 1)	10.10.8.2 (from PC1)	10.10.10.2 (from PC2)
Ping request (traveling from PC1 to PC2) as seen in Wireshark on PC2	f4:0f:1b:a8:52:60 (Router 2)	ac:91:a1:68:b7:ae (from PC2)	10.10.8.2 (from PC1)	10.10.10.2 (from PC2)

- Find out to which devices the mac addresses belong and add the name of the device to the table above.
    - See table above.
- Why are the mac addresses not the same? What happened with these mac addresses at which point in the network?
    - The MAC addresses operate on the Data Link layer (Layer 2) and are only used for local, hop-to-hop communication within the same physical network segment. When a packet reaches a router, the router removes the old Layer 2 Ethernet frame, this is de-encapsulation. To forward the packet to the next segment, the router creates a completely new Layer 2 frame, called encapsulation. In this new frame, the router sets its own interface MAC address as the Source MAC, and the next device's MAC address as the Destination MAC. This is why the MAC addresses are changed at every hop.
- Why are the ip addresses the same?
    - The IP addresses operate at the Network layer (Layer 3) and are responsible for the device to device communication across multiple networks. The Source IP address identifies the original sender of the packet, and the Destination IP address identifies the final target. While routers read the Destination IP address to determine the correct path using their routing tables, they do not modify the original IP packet header during forwarding. Therefore, the Source and Destination IP addresses remain identical throughout the entire process.

Demo

After you did the lab, you should be able to demonstrate the following. Think about how you can demonstrate these achievements.

- Router name
- Network connectivity between PC1 and Router1 over IPv4
- IPv6 SLAAC address on PC1
- Network connectivity between PC1 and Router1 over IPv6
- Network connectivity between PC0 and PC1 over IPv4
- Explain the routers routing table
- Source and Destination MAC and IP address in Wireshark capture on PC0 and PC1